ENEE452 HW10 : networking on micros -attempting to implement a MQTT system with the B-L475E-IOT01A

I have an issue with examples for deploying network software on a micro for something like an MQTT client. The solutions I can find are for particular hardware configurations and are usually wrapped in several hundred megabytes of specialized libraries but use only tiny, deeply buried fractions of the library code to do their work.

It's not the word 'library' that puts me off, its the word 'specialized'. Typically this means that after I've invested the effort to figure out how to get things to work for a particular system, that knowledge isn't useful if I want to port my code to a different system if there are even tiny differences in the new system's hardware components or the development tools being used.

Homework 10 is an initial attempt to address the specialization problem.
What are we trying to do?
Build a test platform.

What is testing going to show us?
(example) How the micro makes a socket call.
(example) What the components of an MQTT conversation with a server are.

Where do we start?
<u>Problem A:</u> build a toy MQTT server and toy MQTT client based on nweb.c and client.c that runs on WSL/macOS using standard UNIX-based network libraries and demonstrates toy MQTT publish and subscribe operations. We are only concerned modeling MQTT and not the networking aspects so your client and server can both be deployed on localhost.

<u>Problem B:</u> create a *translation layer* for the Inventek ISM43362 radio on the B-L475E-IOT01A that <u>lets the micro make the socket calls</u> being used in client.c to connect with the original nweb.c server. The connection here is over wifi so using your PC's hotspot will be the easiest way to model a realworld connection.

If you were in class Tuesday then you know which problem (A or B) your assignment is.
If you missed class on Tuesday then your assignment is Problem A if you last name starts with a letter between 'A' and 'K', else if your last name starts with letters between 'L' and 'Z' your assignment is Problem B.